MADHUPRIYA MAHALINGAN

Senior Software Test Engineer | Test Automation | SAFe Scrum Master | Open to Remote Roles

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Availability: 5 AM – 11 PM IST (ANZ / APAC / EMEA / US-ET overlap friendly)

PROFILE SUMMARY

Software Test Engineer with 6.5 years of experience in automotive functional safety and Hardware-in-Loop (HIL) testing, delivering ASPICE- and ISO 26262-compliant test solutions for infotainment and appliance control systems. Skilled in test automation using Python, Selenium, Playwright, and Robot Framework, with additional experience as a SAFe Scrum Master, working alongside testing responsibilities. Currently expanding automation expertise through hands-on Python test automation projects during a career break.

EXPERIENCE

Bosch Global Software Technologies **Jul 2018 – Jan 2025**

Functional Safety Senior Test Engineer & SAFe Scrum Master — Automotive Infotainment & Appliance Systems

- Tested Access Control Units (ACUs) for home appliances (cooking ranges) using Hardware-in-Loop (HIL) simulation, validating sensor and load conditions with NI devices.
- Performed functional safety testing (ISO 26262, ASIL A/B) for car infotainment systems, automating test execution and reporting using Robot Framework with Python.
- Designed reusable automated test scripts for safety-critical requirements, reducing manual testing effort by 40%.
- Worked throughout in adherence to ASPICE standards (Level 2 and 3), ensuring traceability of requirements in DOORS and IBM RTC.
- Integrated automated test execution into CI/CD workflows using Jenkins, supporting continuous testing across software releases.
- Served as SAFe Scrum Master (Oct 2023 – Jan 2025) alongside testing — facilitated Iteration Planning, daily stand-ups, Iteration Reviews, and Iteration Retrospectives, and took part in PI Planning and Inspect and Adapt events within the Agile Release Train; coached the team on Agile practices while continuing hands-on testing responsibilities.
- Improved test coverage by 30% through comprehensive test planning and automation strategy; improved defect resolution time by 20% by streamlining tracking in JIRA and IBM RTC.
- Served as SAFe Scrum Master for an AI/ML-focused automotive automation team, facilitating Agile delivery in a machine-learning-driven engineering environment.
- Collaborated directly with GM stakeholders and cross-functional product teams to close requirement gaps and resolve critical issues.

Career Break (Jan 2025 – Present): Maternity leave. Used this time to actively upskill — completed a Python Automation Testing certification (Selenium, Playwright, Behave BDD, Robot Framework, Pytest) and built 3 hands-on automation projects (see below).

AUTOMATION PROJECTS

GitHub: github.com/madhu664/Test_Automation_Projects | Bosch project details are kept high level due to confidentiality

- **Functional Safety Test Automation — Bosch:** Automated test execution and reporting for ISO 26262 (ASIL A/B) functional safety verification of automotive systems using Robot Framework and Python; designed reusable scripts that reduced manual testing effort by 40% and integrated runs into Jenkins CI/CD.
- **GM Car Infotainment Test Automation — Bosch:** Automated testing of car infotainment systems for GM vehicles using Robot Framework and Python, following ASPICE standards (Level 2 and 3) and working directly with GM stakeholders.
- **E-Commerce Test Automation — SauceDemo:** Automated the complete online shopping user flow (login, product browsing, cart, checkout) end-to-end using Playwright.
- **HR Management System Automation — OrangeHRM:** Built a test automation suite using Robot Framework, Python, and Selenium to validate core HR workflows, applying keyword-driven test design.
- **EdTech Platform Testing — GUVI:** Developed automated test scripts using Pytest and Selenium to validate key user flows on an EdTech web application.

EDUCATION

- **Master of Technology in Data Science** — Kumaraguru College of Technology (2018–2020)
- **Bachelor of Engineering in Electronics and Communication** — Kumaraguru College of Technology (2014–2018)

CERTIFICATIONS

- ISTQB Automotive Test Specialist (AuT)
- ISTQB Foundation Level
- Professional Scrum Master I (PSM1) — Scrum.org
- SAFe® Scrum Master (expired — renewal in progress)
- Python Automation Testing — GUVI | HCL

TOOLS & SKILLS

- **Automation & Languages:** Python, Selenium, Playwright, Robot Framework, Pytest, Behave (BDD), MySQL
- **Testing Domains:** Functional Safety (ISO 26262), ASPICE, HIL Testing, Infotainment Testing, Regression, Sanity, System-level Testing
- **Tools:** MS Project, IBM RTC/RQM, JIRA, DOORS, Git, Gerrit, Jenkins, Confluence, Collaborator Review Tool, Vector CANoe, NI HIL Systems
- **Agile & Leadership:** SAFe, Scrum, Agile Coaching, Iteration Planning, PI Planning, Inspect and Adapt, Agile Release Train (ART), Team Mentorship, Stakeholder Communication

LANGUAGES

English (Full Professional) • Tamil (Full Professional) • Kannada (Limited Working)